ML Week-06 CS-01

Wednesday, July 8, 2026 7:57 PM

K-Means clustering

K-means → clustering Algorithm

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Unsupervised ML

↙

similar → Group

Use case :

- Customer Segmentation – Group similar customers.
- Product Recommendation – Recommend items to similar users.
- Image Segmentation – Divide an image into regions.
- Document Grouping – Group similar documents.
- Fraud Detection – Find unusual transactions.
- Healthcare – Group patients with similar symptoms.
- Social Networks – Find user communities.
- Market Analysis – Identify customer buying patterns.
- Anomaly Detection – Detect abnormal data points.
- Location Analysis – Group nearby locations.

2D

3D / 4D / ND

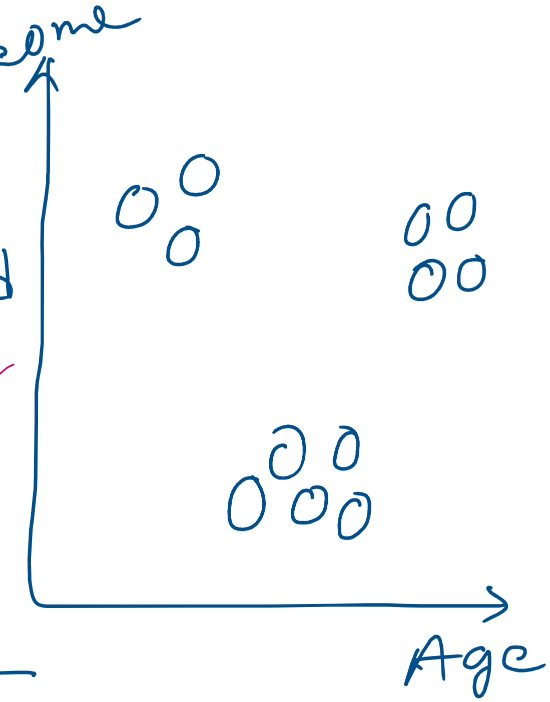
• Location Analysis – Group nearby locations.

3D / 4D / ND

Steps

1) Decide n clusters
2) Initialize centroid
3) Assign clusters
4) Move centroid
5) Finished

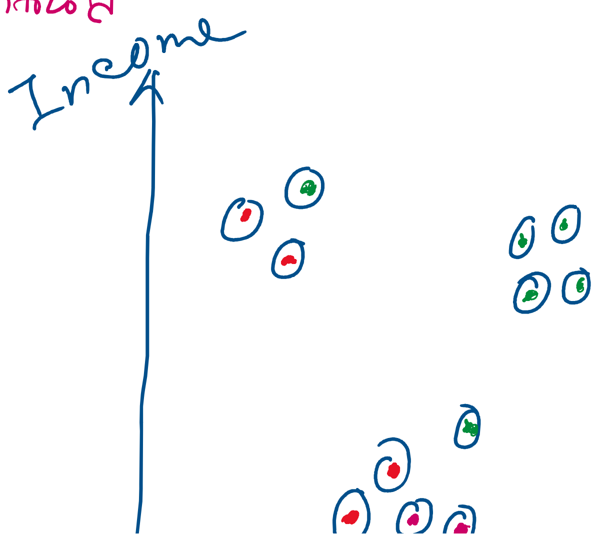
K=3



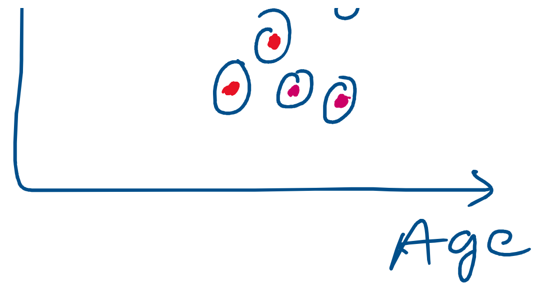
Customer	Age	Income ($)
A	22	25
B	24	28
C	26	30
D	28	32
E	45	80
F	47	85
G	50	90
H	52	95

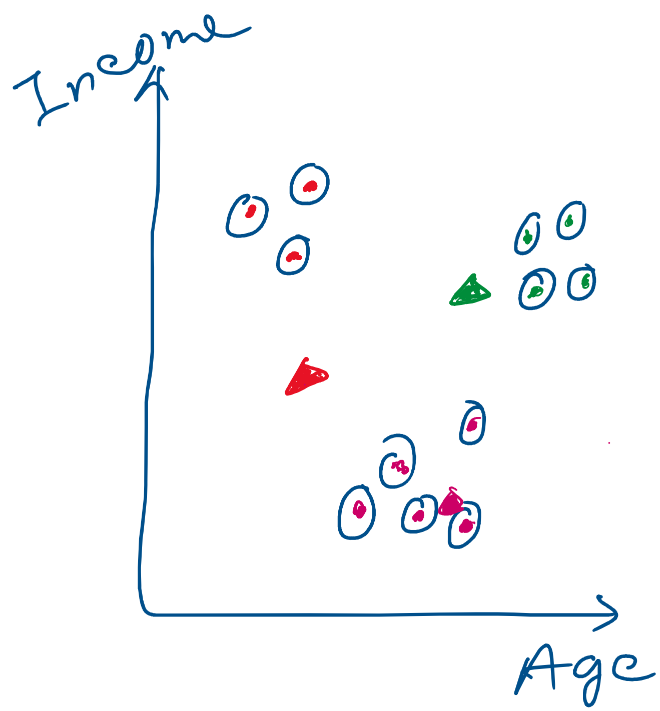
if

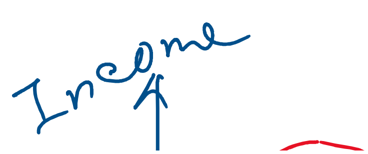
recent position centroid == previous position centroid

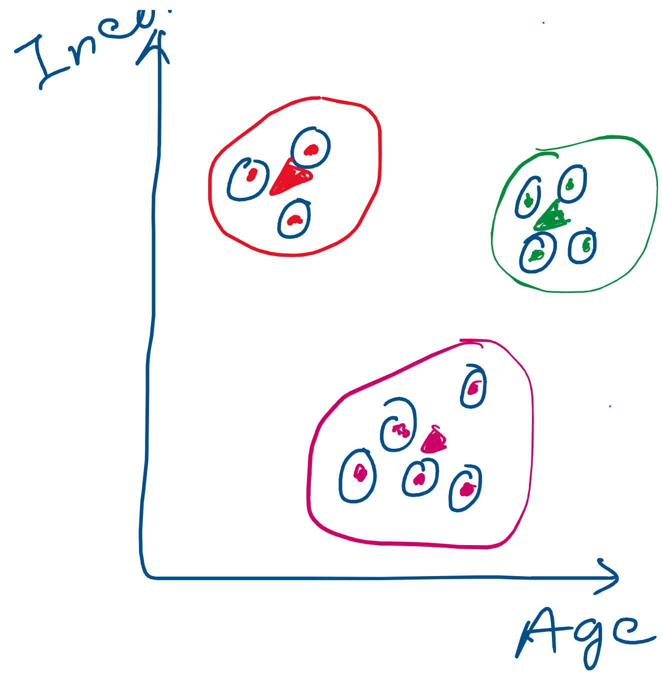


Income (mean)
Age (mean)
↓
point


Age


Income
Age


Income



How can we know number of centroid

Elbow method

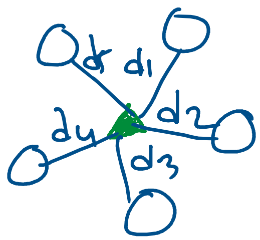
k = ?

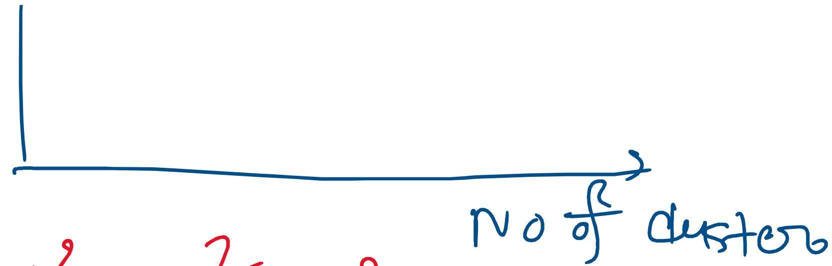
WCSS → within-cluster sum of square

Elbow curve

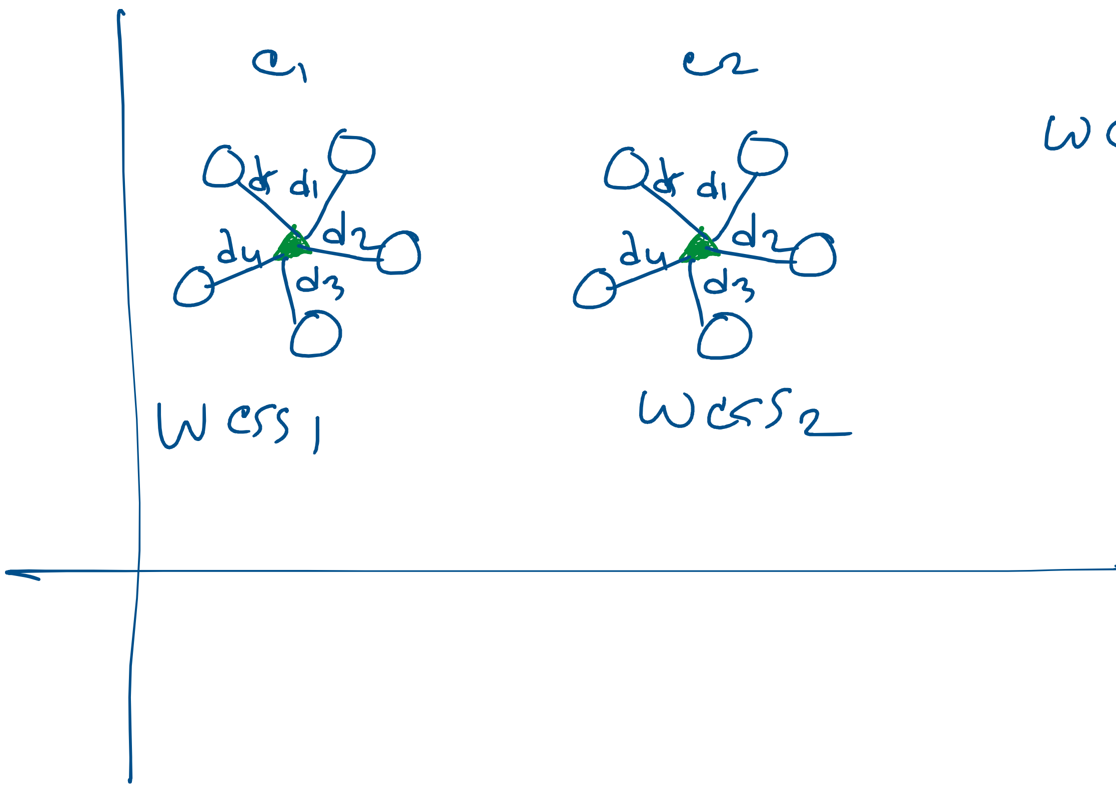
WCSS ?

Qd di?

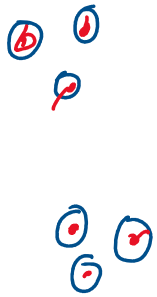




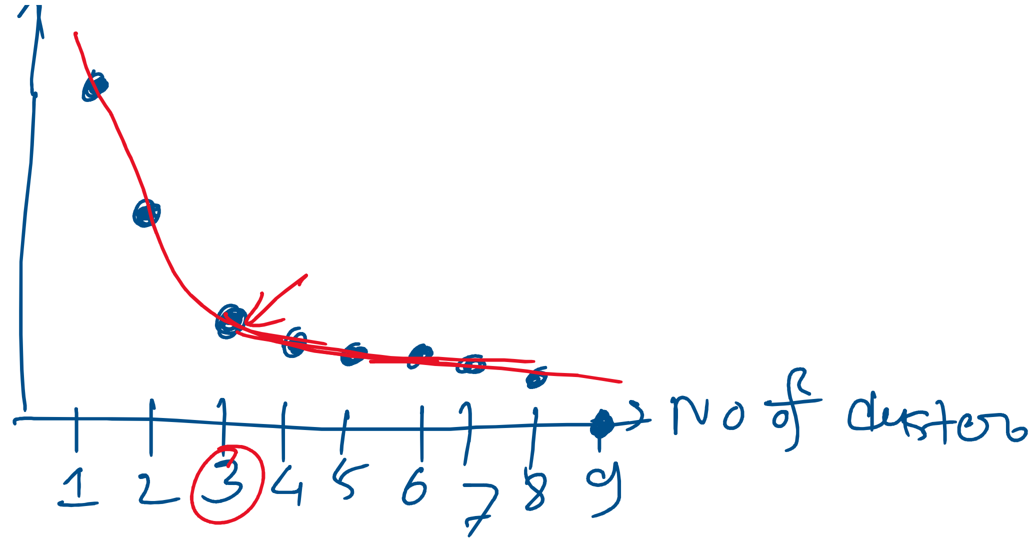
$$WCSS = d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$



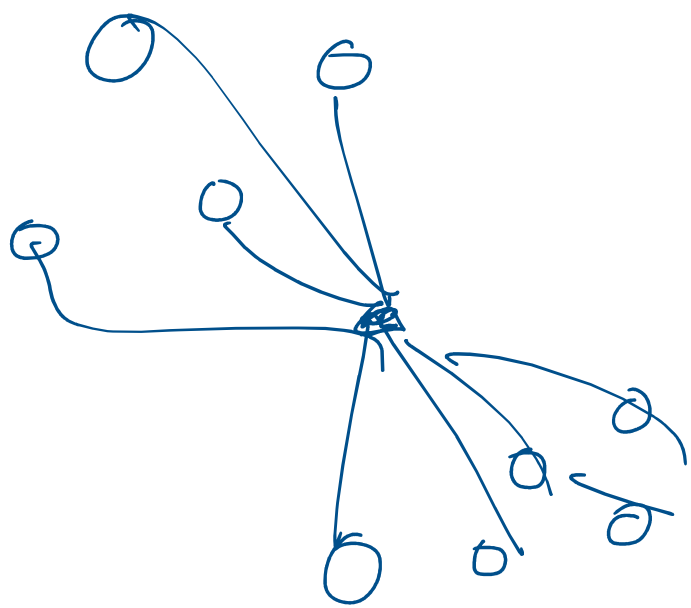
$$WCSS = WCSS_1 + WCSS_2$$



WCSS = 0



$1 \rightarrow WCSS > 2 \rightarrow WCSS > 3 \rightarrow WCSS$



K = 1

Mean